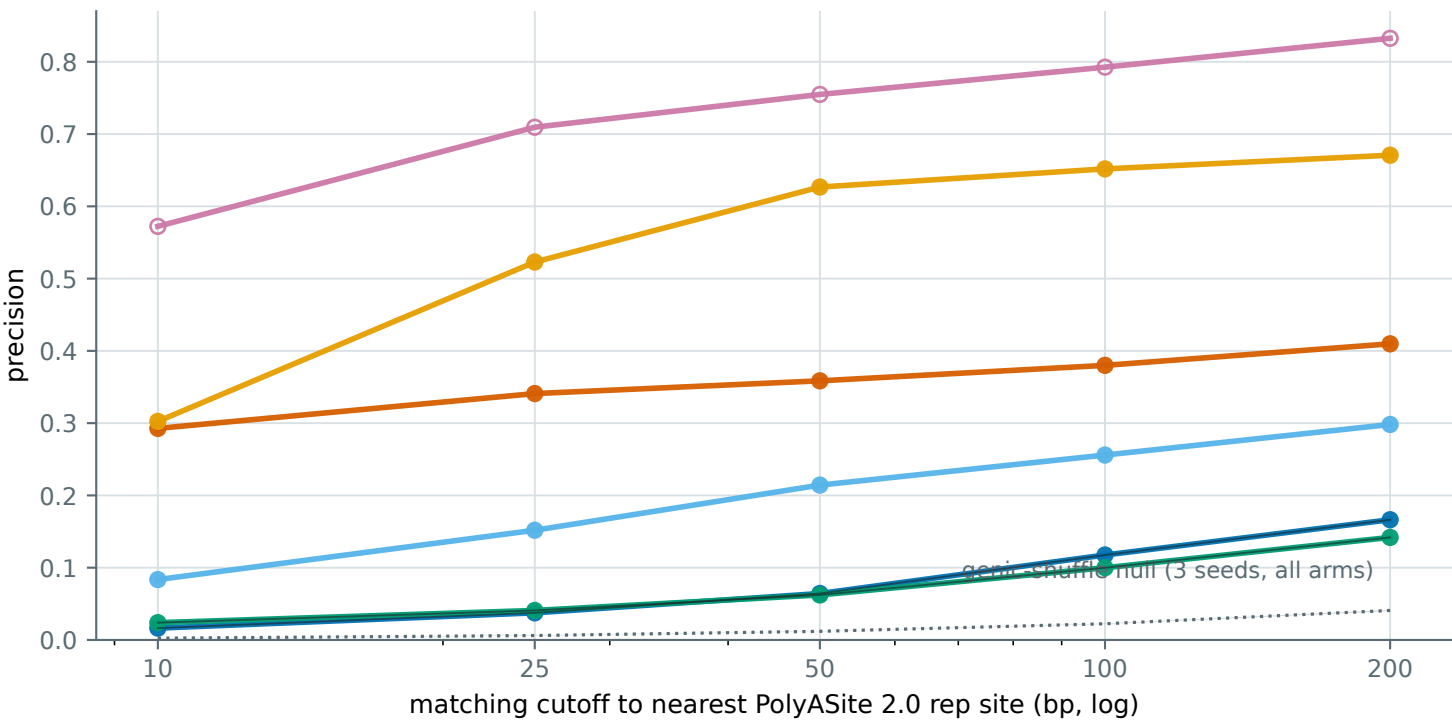


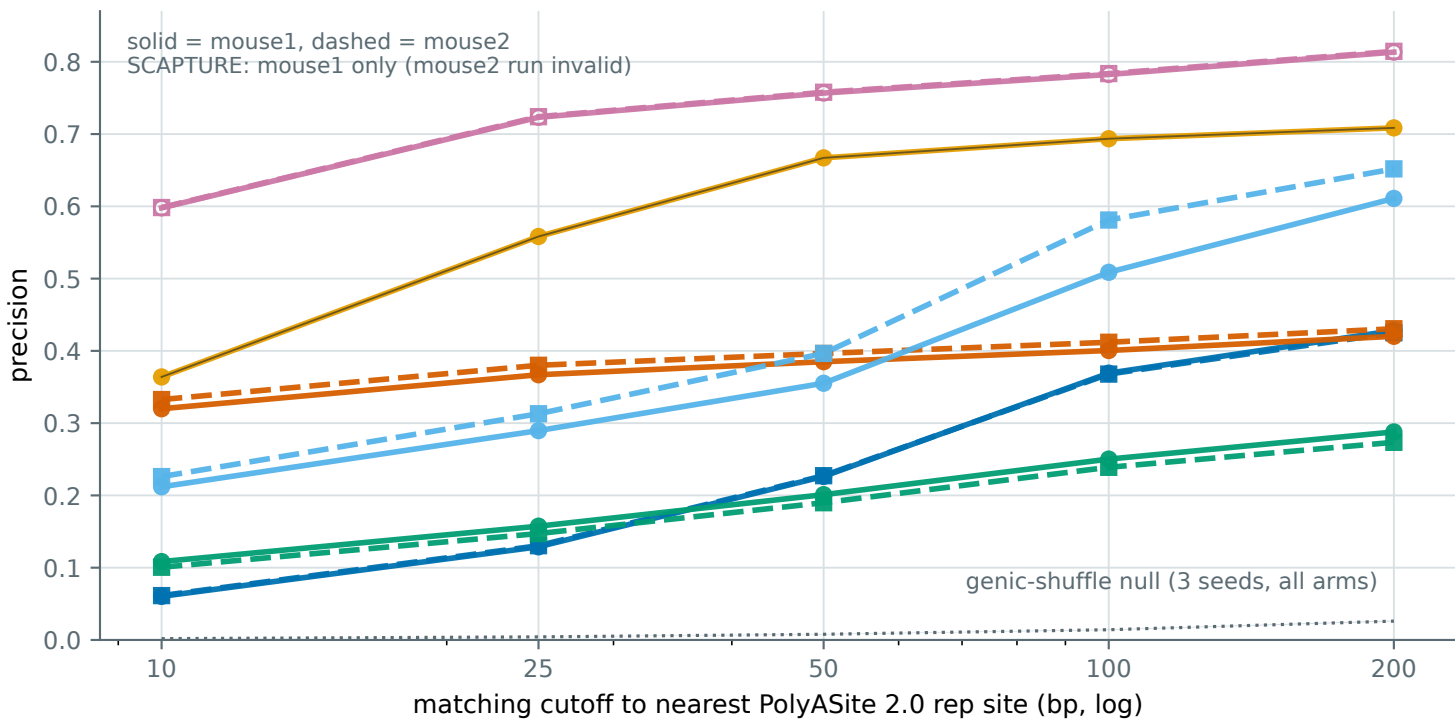
Six PAS callers, two datasets: a trade surface, not a ranking — and PeakATail is last of five de novo tools on human PBMC

SCAPTURE is the precision arm on BOTH datasets (P@100 0.65 human / 0.69 mouse); polyApipe takes the best de novo F1 on both (0.261 / 0.308-0.312) but 62% of its calls are depth-1 singletons that reproduce across replicates only 0.31 of the time; PeakATail is 5th of 5 on PBMC (F1 0.134, P@100 0.118) and 2nd of 5 on testis (0.258-0.260). Every rank order changes between the two datasets — no single accuracy number describes any tool in this field.

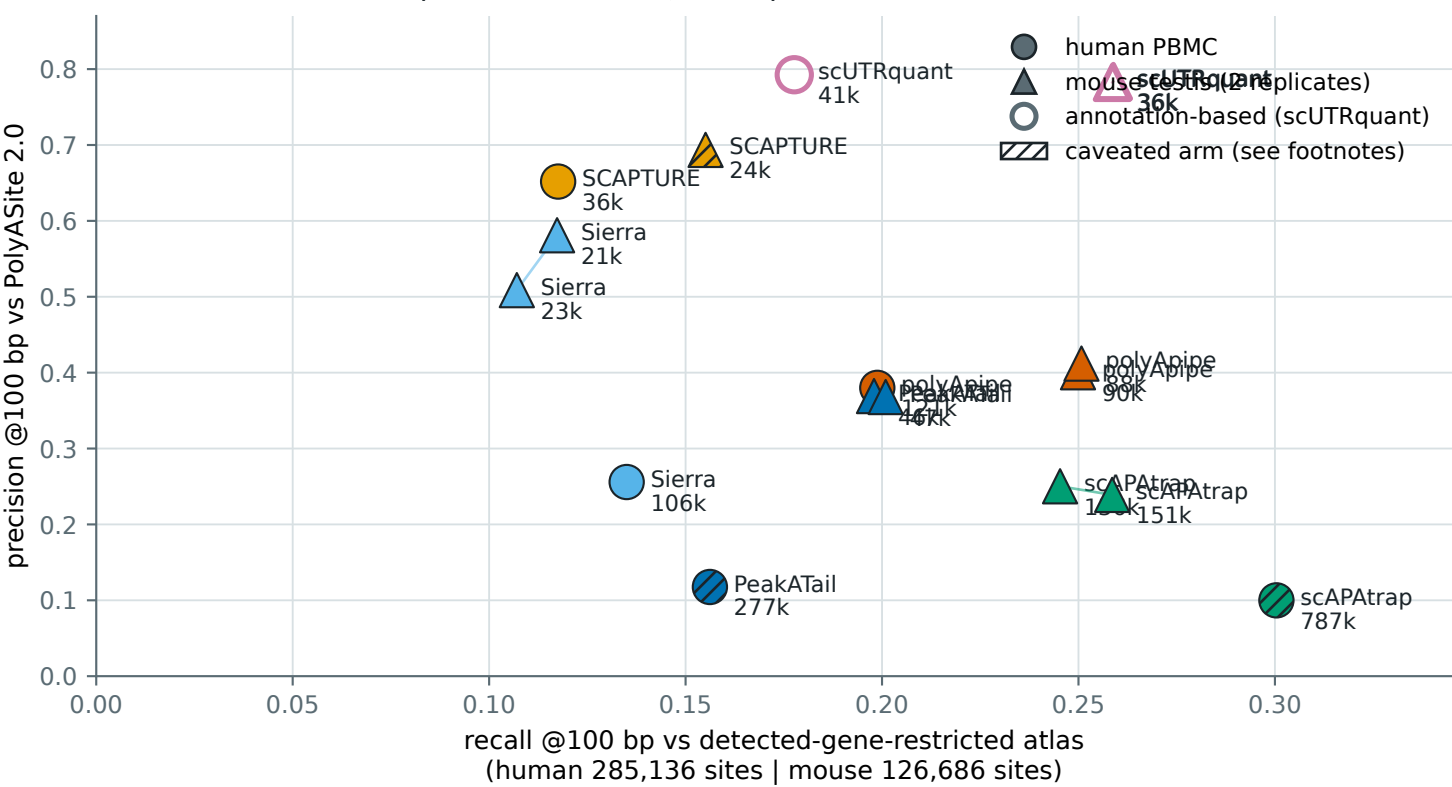
(a) precision vs cutoff — human PBMC (10x pbmc_10k_v3)



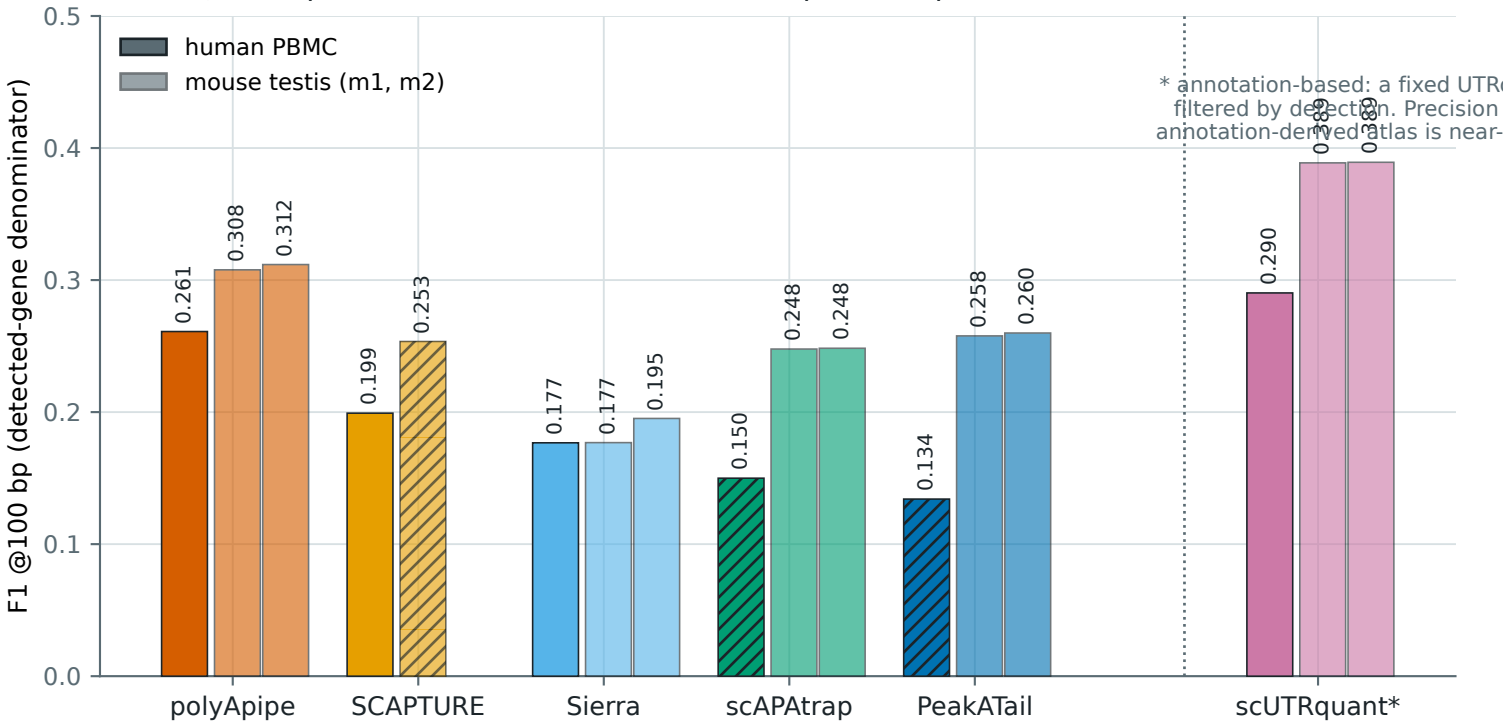
(a) precision vs cutoff — mouse testis (GSE104556)



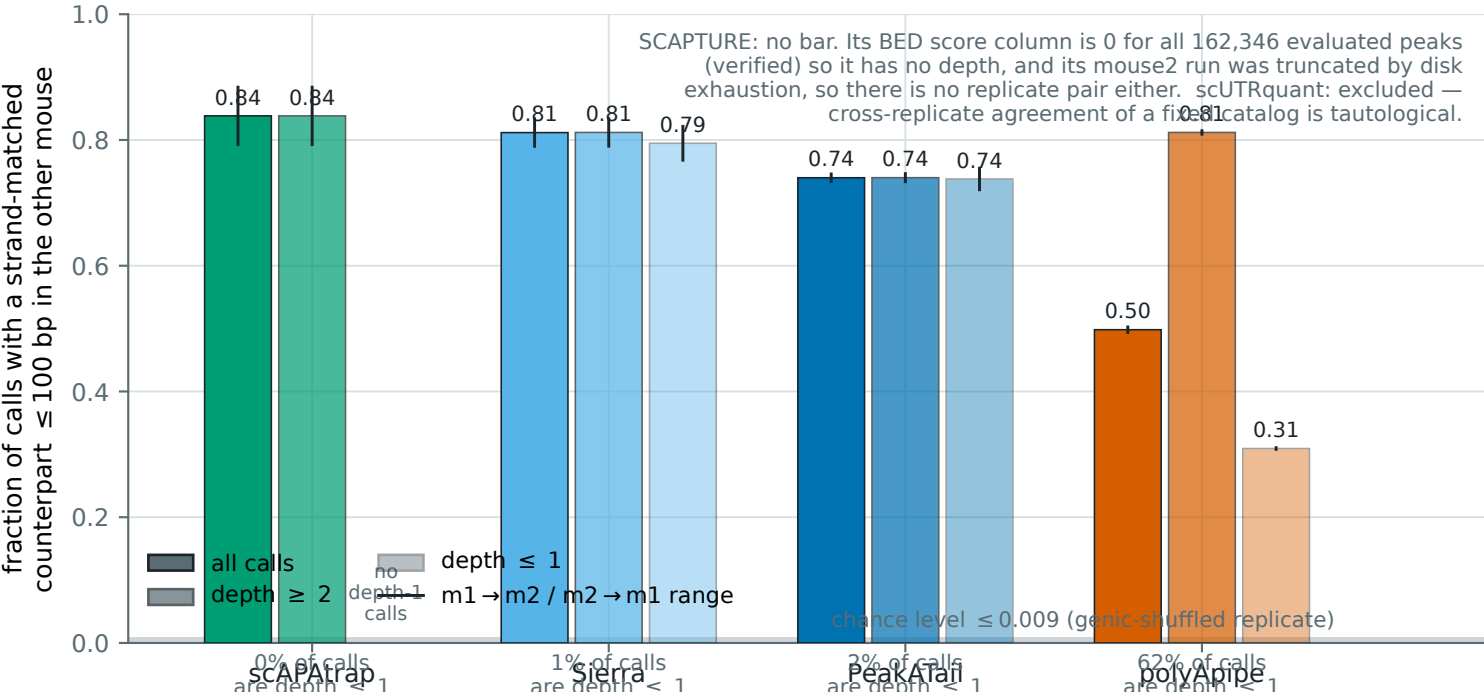
(b) the trade surface: precision-recall @100 bp, labelled with n called



(c) F1 @100 bp — de novo tools ranked; scUTRquant* separated, not ranked



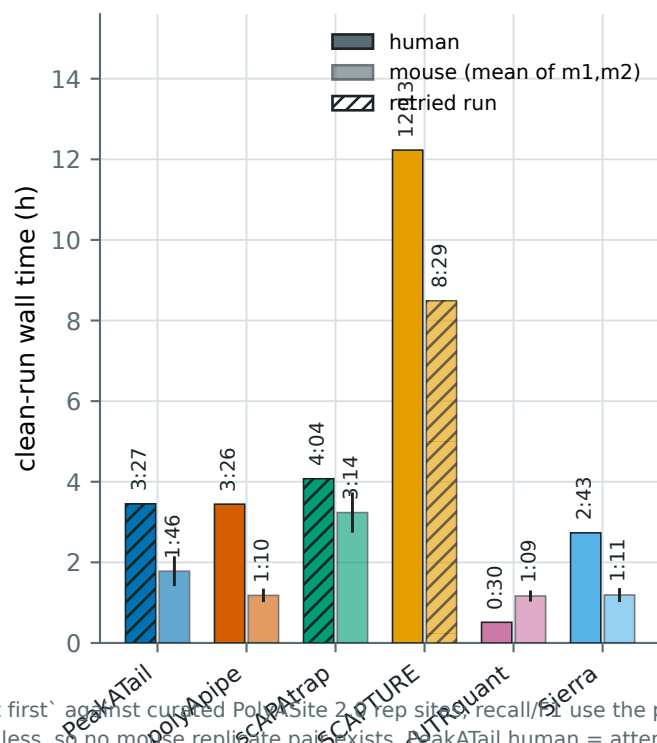
(d) replicate reproducibility, mouse testis — and what depth it rests on



SCAPTURE: no bar. Its BED score column is 0 for all 162,346 evaluated peaks (verified) so it has no depth, and its mouse2 run was truncated by disk exhaustion, so there is no replicate pair either. scUTRquant: excluded — cross-replicate agreement of a fixed catalog is tautological.

SCORING One path for all arms (scripts/benchmark_tools/score_tool.py): 1-bp inferred cleavage points, strand-matched 'bedtools closest -s -d -t first' against curated PolyASite 2.0 rep site. recall/P@ use the per-dataset detected-gene-restricted atlas (human 285,136 sites / 54,851 genes; mouse 126,686 sites / 14,851 genes). CAVEATED ARMS (hatched) SCAPTURE mouse = mouse1 only; the mouse2 run was truncated by disk exhaustion and SCAPTURE exited 0 regardless, so no mouse replicate pair exists. PeakATail human = attempt 4 of 4; attempts 1-3 (1:13:21, 1:52:42, 4:52:58) died on three CellRanger-inputs. PANEL d Depth is per-tool and NOT interconvertible: polyApipe = poly(A)-read peakdepth, PeakATail/Sierra/scAPAttrap = total UMIs. scAPAttrap has no depth≤1 bar because its own reducePeaks(min.cells=10, min.count=10) filter removes them before it reports a peak — its high concordance READ WITH CARE scUTRquant's numbers are near-tautological and are never ranked with the de novo tools. n called differs up to 35-fold across tools (21k-787k), so precision and recall must always be read together with panel (b). SCAPTURE is annotation-guided (peaks are called inside genes).

(e1) wall time



(e2) peak memory

